

p -Affine Surface Areas Generate Centro-Affine Curvature Integrals

A full positive answer to the integral-generation conjecture in arXiv:1208.0783

Automated mathematics run `fa_banach_001`, lane 14

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Status. Candidate full solution, likely valid, pending human review. The theorem answers the precise conjecture concerning integrals of continuous functions of positive centro-affine curvature.

1 Source conjecture

Let $\mathcal{K}_{+,0}^n$ denote the class of C_+^2 convex bodies in $\mathbb{R}^n$ containing the origin in their interior. For $K \in \mathcal{K}_{+,0}^n$, write κ_0 for its centro-affine curvature and $d\mu_K$ for cone measure on ∂K . Then κ_0 is continuous and strictly positive, and

$$\mu_K(\partial K) = n \operatorname{Vol}(K). \quad (1)$$

The source [1] uses, for $p \in \mathbb{R} \setminus \{-n\}$,

$$\Omega_p(K) = \int_{\partial K} \kappa_0^{\frac{p}{n+p}} d\mu_K. \quad (2)$$

On p. 12 it conjectures that the p -affine surface areas, algebraic operations, and closure generate every invariant of the form

$$I_\phi(K) = \int_{\partial K} \phi(\kappa_0) d\mu_K. \quad (3)$$

The complete source paragraph is reproduced here.

and, using (19), we complete the proof of the theorem. □

While it is known that integrals of the form $\int_{\partial K} \phi(\kappa_0) d\mu_K$ are $SL(n)$ -invariant, see also [15], [17], considering the results in [29], and others, including for example the next theorem, we conjecture that the set of p -affine surface areas, with algebraic operations, can generate, by taking the *closure*, all integrals of the above form.

Figure 1: The integral-generation conjecture on p. 12 of the source PDF.

Because the source does not specify a topology for the closure, we prove both pointwise closure on all of $\mathcal{K}_{+,0}^n$ and uniform closure on every natural curvature/volume-controlled stratum.

2 Proof intuition

Formula (2) says that p -affine surface areas are moments of κ_0 . The parameter change $t = p/(n + p)$ attains every real exponent except 1. In particular, choosing

$$p_k = -\frac{2kn}{2k-1}$$

makes $t = 2k$. Hence a finite linear combination of the corresponding Ω_{p_k} is exactly the integral of a polynomial in κ_0^2 .

Every continuous ϕ on $(0, \infty)$ can be written as $\phi(x) = \psi(x^2)$ with $\psi(y) = \phi(\sqrt{y})$ continuous. Approximate ψ locally uniformly by ordinary polynomials. Each fixed positive continuous curvature function has compact range inside $(0, \infty)$, so the polynomial approximation can be integrated. This already uses only the linear span of the p -affine surface areas; products of invariants are not needed.

3 Main theorem

For $a, b, V > 0$ with $a \leq b$, define

$$\mathcal{K}_{+,0}^n(a, b, V) = \{K \in \mathcal{K}_{+,0}^n : a \leq \kappa_0 \leq b \text{ on } \partial K, \text{ Vol}(K) \leq V\}.$$

Theorem 1 (Generation by a fixed countable parameter set). *Let $n \geq 2$ and $\phi \in C((0, \infty))$. Put*

$$p_0 = 0, \quad p_k = -\frac{2kn}{2k-1} \quad (k \geq 1).$$

There are finite linear combinations

$$F_m = \sum_{k=0}^{d_m} c_{m,k} \Omega_{p_k},$$

whose coefficients depend only on ϕ and m , such that

$$F_m(K) \longrightarrow I_\phi(K) \quad \text{for every } K \in \mathcal{K}_{+,0}^n. \quad (4)$$

Moreover, for every $0 < a \leq b < \infty$ and $V < \infty$, the convergence is uniform on $\mathcal{K}_{+,0}^n(a, b, V)$.

Proof. Set $\psi(y) = \phi(\sqrt{y})$ for $y > 0$. For every integer $m \geq 1$, the classical polynomial approximation theorem [2, Chapter 7] gives a polynomial Q_m such that

$$\sup_{y \in [m^{-2}, m^2]} |Q_m(y) - \psi(y)| < \frac{1}{m}. \quad (5)$$

Thus $Q_m(x^2) \rightarrow \phi(x)$ locally uniformly for $x \in (0, \infty)$.

Write $Q_m(y) = \sum_{k=0}^{d_m} c_{m,k} y^k$. For $k \geq 1$, direct algebra gives

$$p_k < -n, \quad \frac{p_k}{n + p_k} = 2k. \quad (6)$$

Thus p_k avoids the sole singular parameter $-n$, and (2) gives

$$\Omega_{p_k}(K) = \int_{\partial K} \kappa_0^{2k} d\mu_K.$$

For $k = 0$, $p_0 = 0$ and $\Omega_0(K) = \int_{\partial K} 1 d\mu_K$. Consequently,

$$F_m(K) := \sum_{k=0}^{d_m} c_{m,k} \Omega_{p_k}(K) \quad (7)$$

$$= \int_{\partial K} Q_m(\kappa_0^2) d\mu_K. \quad (8)$$

Fix $K \in \mathcal{K}_{+,0}^n$. Compactness of ∂K and strict positivity of κ_0 give constants $0 < a_K \leq b_K < \infty$ with $a_K \leq \kappa_0 \leq b_K$. Local uniform convergence therefore implies

$$|F_m(K) - I_\phi(K)| \leq n \text{Vol}(K) \sup_{x \in [a_K, b_K]} |Q_m(x^2) - \phi(x)| \longrightarrow 0,$$

where we used (1). This proves (4).

If $K \in \mathcal{K}_{+,0}^n(a, b, V)$, the same estimate is bounded by

$$nV \sup_{x \in [a, b]} |Q_m(x^2) - \phi(x)|,$$

which tends to zero independently of K . This proves the uniform assertion. $\square$

Corollary 1 (Normalized form). *Let*

$$\widehat{\Omega}_p(K) = \frac{\Omega_p(K)}{\Omega_0(K)}, \quad \widehat{I}_\phi(K) = \frac{I_\phi(K)}{\Omega_0(K)}.$$

Then

$$\sum_{k=0}^{d_m} c_{m,k} \widehat{\Omega}_{p_k}(K) \longrightarrow \widehat{I}_\phi(K)$$

pointwise on $\mathcal{K}_{+,0}^n$, and uniformly over all K satisfying only $a \leq \kappa_0 \leq b$, with no volume bound.

Proof. The normalized cone measure $d\nu_K = d\mu_K / \Omega_0(K)$ is a probability measure. Divide (8) by $\Omega_0(K)$; the approximation error is then at most $\sup_{[a,b]} |Q_m(x^2) - \phi(x)|$. $\square$

4 Consequences, verification, and scope

The theorem places every I_ϕ in the pointwise closure of the linear span of $\{\Omega_{p_k} : k \geq 0\}$ and hence in the closure of the algebra generated by all p -affine surface areas. The parameters form a fixed countable set and satisfy $p_k \uparrow -n$ from below. The normalized corollary directly matches the normalized wording in the source abstract.

The exact-arithmetic script `verify_parameter_map.py` checks (6) for 570 pairs. It is a bookkeeping check, not a substitute for the proof. The negative parameters are harmless because κ_0 is positive and bounded away from zero.

The four run indexes and two bounded web-search rounds used the exact conjecture wording and close mathematical variants. No matching resolution was found; novelty confidence is moderate because the search was not exhaustive.

The result answers the precise p. 12 conjecture (3), not the abstract's broader informal phrasing about arbitrary $SL(n)$ -invariants. Review should focus on the intended closure topology and the admitted range $p \in \mathbb{R} \setminus \{-n\}$.

References

- [1] A. Stancu, *Some Affine Invariants Revisited*, in *Asymptotic Geometric Analysis*, Fields Institute Communications 68 (2013), 343–359; arXiv:1208.0783.
- [2] W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., McGraw–Hill, 1976, Chapter 7.